12.3 What Does "Random" Mean?

WORKSHEET

The question

STEP III 2022, Question 12

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

? **Ask yourself** If the angle AOB is θ, what is the length of the chord AB? (Drop a perpendicular from O.)

STAGE 1 Two points on the circle

Covers: Part (i)

? **Ask yourself** If the angle AOB is θ, drop a perpendicular from O to find AB. The angle θ is uniform on $[0, 2\pi]$: integrate.

STAGE 2 A random point inside

Covers: Part (ii), the density of R

Ask yourself $\mathrm{P}(R \leq r)$ is a ratio of two areas. Differentiate to get the density.

Ask yourself A chord through C makes angle t with OC. How far is the chord from O? So how long is it?

STAGE 3 Averaging over C

Covers: Part (ii), finding $\mathrm{L}(t)$

Pause and try Integrate the chord length against the density of R, with $s = \sin t$ held fixed. Then factorise $1 - \cos^3 t$ and $1 - \cos^2 t$, and use $1 + \cos t = 2\cos^2 \frac{t}{2}$.

STAGE 4 A random direction too

Covers: Part (iii)

Ask yourself T is uniform on $[0, \frac{\pi}{2}]$. Integrate the second form of $\mathrm{L}(t)$. What is the derivative of $\tan \frac{t}{2}$?

What's really going on

Bertrand's three answers

Bertrand's own question was: what is the probability that a random chord is longer than the side of the inscribed equilateral triangle, $\sqrt{3}a$? He gave three methods and three answers.

- Random endpoints: fix one endpoint; the chord is long if the other lies in the middle third of the circle. Probability $\frac{1}{3}$.
- Random radius and a random point on it as the midpoint: the chord is long if its distance from O is less than $\frac{a}{2}$. Probability $\frac{1}{2}$.
- Random midpoint anywhere in the disc: the chord is long if the midpoint lies within $\frac{a}{2}$ of O. Probability $\frac{1}{4}$.

Each method is a different probability distribution on the set of chords, and there is no contradiction: they answer different questions. The lesson is that "uniformly at random" only makes sense relative to a chosen parametrisation, and for continuous sets there is usually no single natural one.

Is there a right answer?

In 1973 the physicist Edwin Jaynes argued that the second method is the "right" one, on the grounds that the answer should not change if the circle is moved or rescaled slightly: a random chord should be like a straw thrown onto a circle drawn on the floor. Only the distribution with uniform distance from the centre has this invariance, and experiments throwing straws confirm $\frac{1}{2}$. The same idea, that the right distribution is the one that respects the symmetries of the situation, underlies *integral geometry*, the mathematics of random lines, which also explains Buffon's needle in Section 12.20.

From paradox to axioms

Bertrand published his chord paradox in *Calcul des probabilités* (1889), partly to mock what he saw as careless uses of "geometric probability". Henri Poincaré discussed it too. In 1900 David Hilbert listed the axiomatisation of probability among his famous problems for the new century, and Andrey Kolmogorov solved it in 1933: probability is a measure on a space of outcomes, and every question about "random" objects must first say which space and which measure. Bertrand's paradox is the standard example of why this is necessary.

Try this next

1. **Bertrand's probabilities.** Verify the three probabilities $\frac{1}{3}$, $\frac{1}{2}$, $\frac{1}{4}$ above.
2. **The other mean.** Show that if the chord's midpoint is chosen uniformly in the disc (the right-hand picture), the expected length is $\frac{4a}{3}$. (*Hint:* this is $L(\frac{\pi}{2})$. Why?)
3. **Random distance.** If the distance of the chord from O is uniform on $[0, a]$, show that the expected length is $\frac{\pi a}{2}$.
4. **Three methods, three distributions.** For each of the three methods, find the probability density of the chord's length ℓ on $[0, 2a]$.

Further reading

- **Edwin Jaynes, "The well-posed problem"**, *Foundations of Physics* 3 (1973). The invariance argument, readable and provocative.
- **Gábor Székely, *Paradoxes in Probability Theory and Mathematical Statistics*** (Reidel). Bertrand's paradox and many others.
- **Daniel Klain & Gian-Carlo Rota, *Introduction to Geometric Probability*** (Cambridge University Press). Integral geometry, for later.

The full guided solution is in Section 12.3:
https://beyondstep.keithmcnulty.org/ch12/bertrand.html